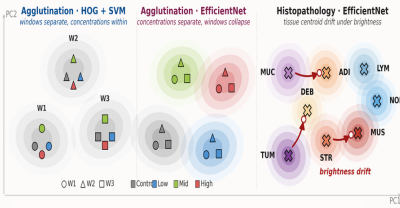


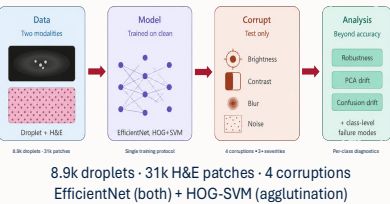
Motivation

- Biomedical images vary by site, instrument, and acquisition day
- Clean accuracy hides class-level fragility under realistic shifts
- PCA on penultimate features surfaces what accuracy misses



Methods

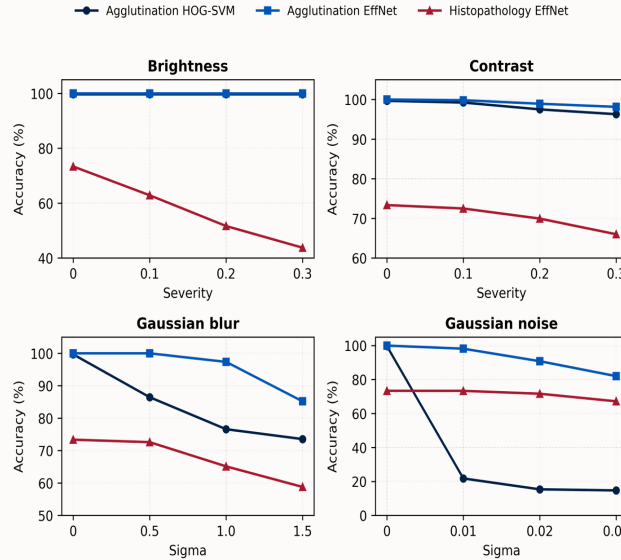
- Two modalities, one protocol
- Train on clean, test on corrupted
- Brightness, contrast, blur, noise
- Robustness + PCA + confusion drift



8.9k droplets · 31k H&E patches · 4 corruptions
EfficientNet (both) + HOG-SVM (agglutination)

Robustness Results

Robustness across agglutination and histopathology image classification

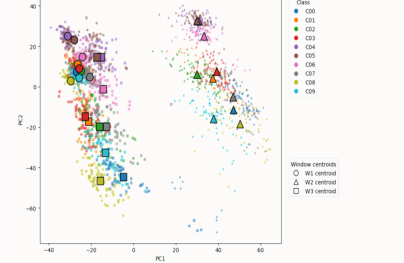


Take-aways:

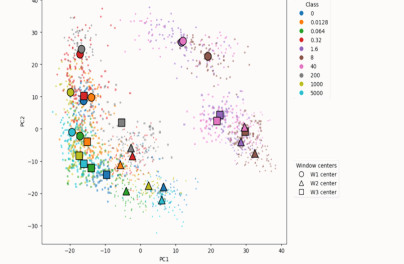
- Same architecture, different modalities, very different corruption profiles
- HOG collapses under noise (~15%);
- EfficientNet stays above 80%
- Histopathology loses ~30 pts to brightness

Agglutination: Information Loss

HOG PCA: Incubation Structure Preserved



EfficientNet PCA: Incubation Structure Merged



Findings:

- Both classify concentration above 98% on clean data
- HOG keeps incubation as a dominant axis (silhouette 0.56)
- EfficientNet learns invariance and discards window structure
- Same accuracy, different preserved information

Representation choice trades invariance for interpretability

Gastric histopathology Corruption Response

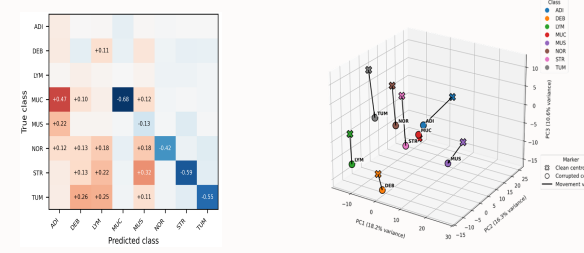
Acronyms:

ADI: Adipose Tissue
DEB: Necrotic Debris
MUC: Mucus
NOR: Normal Mucosa
TUM: Tumor Tissue
LYM: Lymphocyte
MUS: Muscle Tissue
STR: Connective Tissue

References:

- Hendrycks D, Dietterich T. Benchmarking neural network robustness to common corruptions and perturbations. *ICLR2019*.
- Di Salvo F, Doerrich S, Ledig C. MedMNIST-C: comprehensive benchmark and improved classifier robustness by simulating realistic image corruptions. *arXiv 2024*.

Drift Confusion Matrix (Brightness) Centroid Drift in clean PCA space



Findings:

- Corruption pushes class centroids toward their nearest competitor
- Under brightness: MUC → ADI, STR → MUS, TUM → DEB
- Drift directions match the off-diagonal red mass in Drift confusion matrix
- Some classes drift off-axis, flagging alternative failure mechanisms

PCA and drift confusion matrix as paired diagnostic tools for deployment

Key Findings

- Robustness is a representation and data property, not architectural
- Failures are structured and class specific, with predictable confusion directions
- Centroid drift explains dominant confusions geometrically and flags residual classes
- CNNs give invariance; handcrafted features preserve temporal structure

Future Work

- Track newly misclassified samples: boundary effect or content effect?
- Investigate classes where centroid drift misses the confusion target (artifacts, ambiguous tissue, label noise)
- Test stain normalization and targeted augmentation for histopathology;
- HOG and CNN hybrids for agglutination



Take home
message

- Audit representations, not just accuracy
- Use representation drift and confusion drift as joint diagnostics.
- PCA reveals corruption response and information loss
- One method, two case studies, two failure modes



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